

Octopus Chemosensing

Notations

- k_{sub} : number of sub-units in a receptor (typically 5)
- n_{units} : number of possibility/units or number of genes (typically 26)
- $\mathcal{U} = (u_1, u_2, \dots, u_{k_{\text{sub}}}) \in \mathbb{R}^{k_{\text{sub}}} \times \mathbb{R}^{n_{\text{unit}}}$: identity of a detector (homo-pentamers looks like (1,1,1,1,1)) where $u_i \in \mathbb{R}^{n_{\text{unit}}}$
- $r \in \mathbb{R}^{k_{\text{sub}} \times n_{\text{unit}}}$: index / identity of a receptor. Compation of the $\mathcal{U}$
- $\ell \in \mathbb{N}^L$: index / identity of a ligand
- $\mathcal{F}$: ensemble of families of ligand
- f : one specific family of ligand
- $\alpha \in [\text{o}, \text{c}]$: index for open/close or leakiness sometimes.
- $K_{\alpha}^{(r,\ell)}$: dissociation constant of the channel in the α state, between a receptor r and a ligand ℓ
- $\Delta E_{\alpha}^{(r,\ell)} = E_{\alpha}^{(r,\ell)} = E_{\text{bound}} - E_{\text{unbound}}$: energy difference between the bound and unbound state.
- $\epsilon_r = E_c - E_o$: energy difference between the closed and opened state of the channel without ligand. Leakiness = e^{ϵ}
- $\theta = \{(\Delta E_{\alpha}^{(r,\ell)})_{\alpha,\ell}, \epsilon_r\}_r = \{\theta_r\}_r$ compact indexing for the characteristics of a channel
- $\bar{M} = \{\mu_{\alpha}^{(r,f)}\}_{r,f}$ and $\bar{\Sigma} = \{\sigma_{\alpha}^{(r,f)}\}_{r,f}$: respectively the matrix of the mean and std deviation of the distribution from which the energies are drawn for the receptor r in response to the family of ligands f . should the matrices depends on alpha, or should one be a shifted version of the other. In anyway the shift should be the same for every ligands, for a given receptor
- w generic name for $\mu_{\alpha}^{(r,f)}$ and $\sigma_{\alpha}^{(r,f)}$.
- N : total number of receptors in the array.
- $\mathcal{A} = (a_0, a_1, \dots, a_N)$: activity value of the array of receptors.

Vocabulary

- Receptor: a single ion channel.
- Unit: a single protein that constitue a receptor.
- ligand: an arbitrary molecule.
- Pocket: the region of the unit within which a ligand bind.
- Residue: a piece of molecule that composes the pocket or/and a ligand.

Translation from Experimentalist

Experimentalist Term	Physicist / MWC Term	Definition in Context
Efficacy	Response Amplitude (I_{max})	The peak response (current) relative to other receptors.
Potency / Sensitivity	Apparent Affinity (EC_{50})	The ligand concentration required to reach half-maximal activation.
Selectivity / Tuning	Ligand Specificity (K_d)	The breadth of ligands a receptor responds to (e.g., “broad” vs. “narrow”).

Experimentalist Term	Physicist / MWC Term	Definition in Context
Cooperativity	Allosteric Coupling	How subunit interactions (like heteropentamers) affect the opening curve.
Diversity	Mutual Information	The capacity of the receptor array to distinguish between different ligands.

1. Experimental procedure

Our research investigates the generation of functional diversity in octopus olfactory channels, which are understood to be pentamers ($k_{\text{sub}} = 5$) formed from a pool of $n_{\text{unit}} = 26$ different proteins (e.g., ‘CR518’, ‘CR918’, etc.). We hypothesize that an olfactory cell’s response to a ligand is not from a single, defined channel type but rather from a combinatorial ensemble of all possible pentameric stoichiometries and combinatories (e.g., AAAAA, ABBBB, AABBB, ABABB etc..) based on the genes it expresses. We want to **measure the diversity of resposne generated by combining hetero-pentamers**

We will fit the experimental dose-response curves for the homopentamers ‘CR518’ and ‘CR918’, as well as the heteropentameric ‘CR518_918’ combination, across a panel of 11 ligands (including “Medroxyprogesterone”, “Androsterone”, “Progesterone”, etc.). -> Experimental dose-response curves measure the current generated by the diffusion of small ions through a membrane. -> This current is proportional to the opening probability of the channels embedded in the membrane. -> To rule-out the influence of this proportionality factor, we focus our fitting procedure on the renormalize current:

$$\text{renormalized current} = \frac{I(c) - I_{\min}}{I_{\max} - I_{\min}} = \frac{p_{\text{open}}(c) - p_{\min}}{p_{\max} - p_{\min}}$$

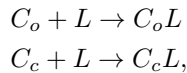
2. Single Receptor Characteristics

2.1 MWC Model

Each receptor is made of k_{sub} sub-units among n_{unit} possible units. We denote $\mathbf{r} = (r_0 \dots r_{k_{\text{sub}}}) \in \mathbb{R}^{n_{\text{unit}} \times k_{\text{sub}}}$ the identity of a receptor characterized by the identity of each of the units that composes it. The response of a receptor $\mathbf{r}$ to a ligand ℓ at concentration c can be written using the MWC model for ion channel opening:

$$p_o^r(c, \ell) = \frac{\prod_{u \in \mathcal{U}} (1 + c/K_o^{(u, \ell)})}{\prod_{u \in \mathcal{U}} (1 + c/K_o^{(u, \ell)}) + e^{-\epsilon_u} \prod_{u \in \mathcal{U}} (1 + c/K_c^{(u, \ell)})}$$

Where the K ’s are called the affinities. ### 2.2 Microscopic model of a receptor To make the link between the dissociation constant, we consider the chemical reactions:



where C_o and C_c are respectively the channel in the open and closed state, L is the ligand, and $C_o L$ and $C_c L$ are the ligand-channel dimers. The the dissociation constants reads :

$$\begin{aligned} K_o &= \frac{[C_o][L]}{[C_o L]} \\ K_c &= \frac{[C_c][L]}{[C_c L]}, \end{aligned}$$

where $[C_o]$ and $[C_c]$ are respectively the concentration of open and closed channel, whereas, $[L]$ is the concentration of ligand in solution. From a statistical mechanics point of view, we can write for any specie α

$[\alpha] \propto p(\alpha) = 1/Z e^{\beta E_\alpha}$. Where Z is the partition function of the system. As a result, we write the dissociation constants as:

$$K_o^{(r,\ell)} = \exp [\Delta E_o^{r,\ell}]$$

$$K_c^{(r,\ell)} = \exp [\Delta E_c^{r,\ell}],$$

with:

$$\Delta E_o^{(r,\ell)} = E(C_o L) - E(C_o) - E(L)$$

$$\Delta E_c^{(r,\ell)} = E(C_c L) - E(C_c) - E(L)$$

Notice that a good ion channel has $K_c > K_o$ (greater dissociation when closed). With that convention that means $\Delta E_c > \Delta E_o$.

Each affinity correspond to the response of a sub-unit to a ligand. $\epsilon_{\mathcal{U}}$ is the energy difference between the closed and opened state of the channel when there are no ligands, it is thus independent of ℓ . We write it as the simple sum of the contribution of each individual unit:

$$\epsilon_{\mathcal{U}} = \sum_u \epsilon_u$$

2.3 Parameters Degeneracy

in [http://dx.doi.org/10.1021/acs.jpcb.6b12672] supplementary material, p.S25, they look at the hessian matrix:

$$H_{i,j} = \frac{\partial p_o}{\partial x_i \partial x_j}$$

, where $x = \beta(\epsilon, \Delta E_o, \Delta E_c)$. They find that the Hessian has three eigenvalues : $\{2 \times 10^{-2}, 9 \times 10^{-4}, 2 \times 10^{-7}\}$. The last eigen value has a eigen vector $(2, 1, 0)$, and is sensitively smaller than the two others, highlighting a sloopy mode. in the $(2, 1, 0)$ the parameter $e^{-\beta\epsilon/2} K_o$ is constant. As a result, we can rewrite the opening probability assuming that: $\left(e^{\beta\epsilon/2} + e^{\beta\epsilon/2} \frac{c}{K_o}\right)^2 + \left(1 + \frac{c}{K_c}\right)^2 \approx \left(e^{\beta\epsilon/2} \frac{c}{K_o}\right)^2$

$$p_o^{(r,l)}(c) = \frac{\prod_u \left(\frac{c}{\tilde{K}_o^{u,l}}\right)}{\prod_u \left(\frac{c}{\tilde{K}_o^{(u,l)}}\right) + \prod_u \left(1 + \frac{c}{K_c^{(u,l)}}\right)},$$

where $\tilde{K}_o = K_o e^{\beta\epsilon}$

2.4 Adimensional activation curve

We now write the final activation curve by adimensionalizing the concentration using EC_{50} the concentration at half activation. In the limit $1 \ll e^{\beta\epsilon/k_{\text{unit}}} \ll \left(\frac{K_c}{K_o}\right)^{k_{\text{unit}}}$, we have

$$[EC_{50}^{(u,l)}] = \left(\prod_{u=1}^{k_{\text{unit}}} \tilde{K}_o^{(u,l)}\right)^{1/k_{\text{unit}}}.$$

In that case, we can write the opening probability:

$$p_o^{(r,l)}(x) = \frac{x^{k_{\text{unit}}}}{x^{k_{\text{unit}}} + \prod_{u=1}^{k_{\text{unit}}} \left(1 + \frac{x[EC_{50}^{(u,l)}]}{K_c^{(u,l)}}\right)} = \frac{x^{k_{\text{unit}}}}{x^{k_{\text{unit}}} + \prod_{u=1}^{k_{\text{unit}}} (1 + \delta^{(u,l)} x)},$$

where $x = c/$

3. Generating Ligand/Unit interactions : The Environment

-> In general, the microscopic interaction that compose the affinity between a ligand and a receptor is a finite pool (Van der Waals, aromatic interactions, electrostatic interactions, hydrophobic interactions). -> This leads the interaction energies between a given receptor and different ligands to be correlated. For instance, a polar receptor should have affinity for every polar ligands. -> Instead of making a complete, microscopic model of each receptor and ligands based on the specific microscopic interaction, we build affinities values by sorting the response of a given receptor with respect to a family of ligands (eg: acids, alcohol, aldehydes, esters, aromatics, terpenes etc. . .) -> In summarize, this approach accounts for the correlation within a family, and neglect the inter-family correlation. -> Following previous work on drosophilia [10.1038/srep21841] and humang [10.1038/sdata.2015.2], we model the distribution of interaction energy between a family and given receptor as Gaussian. -> This can be interpreted as a consequence of the central limit theorem, where the interaction energy represents an effective average of many microscopic interactions.

-> The physical characteristics of an array of receptor is model through two matrices: $\bar{\Sigma} = \{\mu_\alpha^{(r,f)}\}_{r,f}$ and $\bar{M} = \{\sigma_\alpha^{(r,f)}\}_{r,f}$. Where each entry gives the mean or standard deviation of a given receptor to a family of ligand. Notice that this isn't a squared matrix. -> In the rest, we will optimize these matrices to maximize mutual information between ligands identity/concentration and activity of the array.

On top of that, the concentration distribution of each Ligands is assumed to be Gaussian as well. We draw the average μ_c^ℓ , and standard deviation σ_c^ℓ , each of them drawn from a uniform distribution. we should think about it again

We assume that the receptor array is designed for sensing a large number of ligand. Any ligand is defined by

1. its interaction with every units: $\{E_\alpha^{u,\ell}\}_{\alpha,u}$
2. A probability distribution of concentration: $p_\ell(c)$

To model the random encounter of a ligand, we categorize them into families. All the ligands of a given family is assumed to have similar properties. As a consequence, the random encounter of a ligand of a given family lies within a fairly narrow distribution (Normal) specific to each sub-unit. For a given family f :

$$P_\alpha^{u,f}(E) = P(E|\alpha, u, f) = \mathcal{N}(\mu_\alpha^{u,f}, \sigma_\alpha^{u,f}) = \sigma_\alpha^{u,f} \mathcal{N}(0, 1) + \mu_\alpha^{u,f}$$

Where $\alpha \in \{o, c\}$. The goal is to optimize the mean and standard deviation of the probability distribution of each individual sub-unit to each family, in order to maximize the mutual information between a ligand's identity and/or its concentration in a given environment.

In total, the number of parameters to optimize for a single receptor is:

$$N_{\text{param}} = \underbrace{\underbrace{n_{\text{unit}}}_{\# \text{ of different units}} \times \underbrace{\left(\underbrace{2}_{\mu, \sigma} \times \underbrace{2}_{o, c} \times \underbrace{N_{\mathcal{F}}}_{\# \text{ of families}} \right)}_{\text{interaction}} + \underbrace{n_\epsilon}_{\# \text{ parameter for } \epsilon}}_{\# \text{ parameter for the concentration}} + \underbrace{N_{\mathcal{F}} \times n_c}_{\# \text{ parameter for the concentration}}$$

4. Information Produced by an Array of Receptors

We now consider an array of N receptors, and write the activity of this array as $\mathcal{A} = (a_0, a_1, \dots, a_N)$. The goal of this section is to design an algorithm to optimize the the mutual information between the activity of the array and ligand's identity and/or concentration for a given environment:

$$I(\mathcal{A}; (c, \ell)) = H(\mathcal{A}) - H(\mathcal{A}|(c, \ell)),$$

We consider the mapping between affinities, and activity as deterministic, noiseless: $H(\mathcal{A}|(c, \ell)) = 0$. Thus we want to maximize

$$H(\mathcal{A}) = - \sum_{\mathcal{A}} p(\mathcal{A}) \log[p(\mathcal{A})] = \mathbb{E} [\log[p(\mathcal{A})]] .$$

4.1 Single Receptor Problem

4.1.1 Kernel Density Estimator Let us first consider a single receptor, and its probability distribution of activation $p(a)$. Starting from the empirical probability distribution:

$$\tilde{p}(a) = \frac{1}{N} \sum_{i=0}^{N_{\text{sample}}} \delta(a_i - a),$$

where N_{sample} is the number of sampled odors. We approximate the real density probability distribution using a Gaussian Kernel:

$$\hat{p}(a) \approx \frac{1}{Nh} \sum_{i=0}^{N_{\text{sample}}} \frac{1}{h\sqrt{2\pi}} \exp\left[-\frac{(a - a_i)^2}{2h^2}\right],$$

where h is the bandwidth of the kernel. Let us start with a simple rule of thumb for h . Assuming that the resulting distribution is somewhat Gaussian, we find:

$$h \approx 1.06\tilde{\sigma}N_{\text{sample}}^{-1/5},$$

where

$$\tilde{\sigma} = \sqrt{\frac{1}{n-1} \sum_{i=0}^{N_{\text{sample}}} (a_i - \bar{a})^2}$$

is the empirical standard deviation of the samples. We can then write the empirical entropy inferred from the distribution of a single receptor:

$$\tilde{H} = -\frac{1}{N_{\text{sample}}} \sum_{i=0}^{N_{\text{sample}}} \log[\hat{p}(a_i)]$$

Notice that $\tilde{H}(a_0, \dots, a_{N_{\text{sample}}})$ is a function of all the sampled activity ##### 4.1.2 Chain Rule For the Gradient

To compute the gradient of the entropy of a single receptor, we use the chaine rule:

$$\frac{\partial \tilde{H}(\{a_i\})}{\partial w} = \sum_{i=0}^{N_{\text{sample}}} \frac{\partial a_i}{\partial w} \cdot \frac{\partial \tilde{H}}{\partial a_i}$$

Where $a_i(c, E_o, E_c, \epsilon)$ is given by the MWC model, and E_o, E_c, ϵ come from a distribution defined in Section 3. Notice that the gradient $\partial_{a_i} \tilde{H}$ can be computed semi-analytically from the inferred $\hat{p}$. Now, writing the derivative of a_i :

$$\frac{\partial a_i}{\partial w} = \frac{\partial a_i}{\partial E_c} \cdot \frac{\partial E_c}{\partial w} + \frac{\partial a_i}{\partial E_o} \cdot \frac{\partial E_o}{\partial w} + \frac{\partial a_i}{\partial \epsilon} \cdot \frac{\partial \epsilon}{\partial w}.$$

In our case, because E_c, E_o, ϵ are independent variable, only one of the derivative is non-zero. Let us assume that $w = \mu_c$:

$$\frac{\partial \tilde{H}}{\partial \mu_c} = \frac{\partial E_c}{\partial \mu_c} \sum_{i=0}^{N_{\text{sample}}} \frac{\partial a_i}{\partial E_c} \cdot \frac{\partial \tilde{H}}{\partial a_i}.$$

Finally, we use the reparametrization trick:

$$E_c = \mu_c + \zeta_i \sigma_c,$$

where $\zeta \sim \mathcal{N}(0, 1)$. In the last expression, everything is easy to compute.

4.3 Whole Array Problem

4.3.1 The problem As dimensionality of the PDF increases, KDE methods becomes increasingly bad. Typically, the number of samples required for a descent estimate of the PDF grows as 10^D for a PDF of dimension D. As a result, we rely on an approximated loss function.

4.3.2 Approximated loss function Our algorithm We expand the entropy using the information expansion:

$$H(\mathcal{A}) = \sum_r H(a^r) - \sum_{r < r'} I(a^r; a^{r'}) + \sum_{r < r' < r''} I(a^r; a^{r'}; a^{r''}) - \dots$$

We first write the mutual information between two receptors activity, **assuming that they are Gaussian**, or not too far from Gaussian.

$$I(a^r; a^{r'}) = \frac{\text{Cov}(a^r, a^{r'})^2}{2 \cdot \text{Var}(a^r) \cdot \text{Var}(a^{r'})}$$

Where $\text{Cov}(a^r, a^{r'}) = \mathbb{E}[a^r - \mathbb{E}[a^r]] \mathbb{E}[a^{r'} - \mathbb{E}[a^{r'}]]$, and $\text{Var}(a^r) = \text{Cov}(a^r, a^r)$.

Thus, neglecting three-receptors correlations, we write the loss function as:

$$\mathcal{L} = \underbrace{- \sum_r H(a^r)}_{\text{optimize individual receptors}} + \lambda \underbrace{\sum_{r < r'} \text{Cov}(a^r, a^{r'})^2}_{\text{make them as different as possible}}$$

Ideally, we would like the receptors to be **independent**, which is the **strong condition**

$$p(a^r, a^{r'}) = p(a^r)p(a^{r'}),$$

In this approximation, we optimize for the receptors to be **uncorrelated** which is the **weak condition**:

$$\text{Cov}(a^r, a^{r'}) = 0$$

If independance implies uncorrelated, uncorrelated, in general, doesn't imply independance. There is only one case for which is the same, and it is for Gaussian distributed variables. Indeed, if $\forall i \neq j \text{ Cov}(a^i, a^j) = 0$, then the covariance matrix Σ is diagonal. The multivariate probability distribution:

$$p(a^r, \dots, a^N) = \frac{1}{(2\pi)^{2/N} \det(\Sigma)} \exp \left[-\frac{1}{2} (\mathcal{A} - \mathbb{E}[\mathcal{A}]) \Sigma^{-1} (\mathcal{A} - \mathbb{E}[\mathcal{A}]) \right],$$

where $\mathcal{A} = (a^r, \dots, a^N)$. Which can be easily separated as a product of the individual distribution.

In the case of our optimization problem, we can write the entropy of a Gaussian variable as:

$$H_{\mathcal{G}}(\mathcal{A}) = \frac{1}{2} \log((2\pi e)^N |\det(\Sigma)|)$$

Maximizing $H_{\mathcal{G}}(\mathcal{A})$ implies maximizing $\det(\Sigma)$ which implies: - Maximizing the diagonal terms: $\text{Var}(a^r)$. - Minimizing the off-diagonal terms: $\text{Cov}(a^r, a^{r'})$. Instead of computing the determinant which is costly, we minimize the covariance terms. This is also called whitening.

4.3.3 Gradient of the Covariance The empirical covariance writes:

$$C_{r,r'} = \frac{1}{N_{\text{sample}} - 1} \sum_{i=0}^{N_{\text{sample}}} (a_i^r - \bar{a}^r)((a_i^{r'} - \bar{a}^{r'})),$$

with the covariance loss being: $\mathcal{L}_{\text{cov}} = \sum_{r < r'} C_{r,r'}^2$, which ends up being a function of $\{a_i^r\}_{\substack{r \in [1, N] \\ i \in [1, N_{\text{sample}}]}}$. Thus we can write the gradient of the covariance loss:

$$\frac{\partial \mathcal{L}_{\text{cov}}}{\partial w} = \sum_{i=1}^{N_{\text{sample}}} \sum_{r=1}^N \frac{\partial \mathcal{L}_{\text{cov}}}{\partial a_i^r} \cdot \frac{\partial a_i^r}{\partial w}.$$

In the single receptor section, we explain how to compute $\frac{\partial a_i^r}{\partial w}$ with the chain rule. Notice that

$$\frac{\partial \mathcal{L}_{\text{cov}}}{\partial a_i^r} = \sum_{r' \neq r} C_{r,r'} (a_i^{r'} - \bar{a}^{r'}),$$

takes the form of This is a restoring force proportional to the correlation $C_{r,r'}$.

5. Fixed concentration problem

We consider an array that respond to a fixed concentration.

1. The Analytical Solution

First, we define the normalized activity $y(c) = p_o(c)/p_{\text{max}}$. The Hill plot maps $\log(c)$ against $\log\left(\frac{y}{1-y}\right)$. The Hill coefficient is the derivative:

$$n_H = \left. \frac{d \log\left(\frac{y}{1-y}\right)}{d \log c} \right|_{c = EC50}$$

If you take your simplified probability equation, compute $y(c)$, and run it through this derivative, a remarkably elegant physical result drops out:

$$n_H = k_{\text{sub}} - \left. \frac{\sum_{u=1}^{k_{\text{sub}}} \theta_u(c) - k_{\text{sub}} f_c(c)}{1 - f_c(c)} \right|_{c = EC50}$$

$\theta_u(c) = \frac{c/K_c^{(u)}}{1+c/K_c^{(u)}}$ is the occupancy of subunit u if the channel were locked in the closed state.

$f_c(c) = \prod_u \theta_u(c)$ is the probability that the channel is fully bound while in the closed state.

References